C7.1 ARCACHE and ARDOMAIN requirements

A cache clean operation is used to ensure that a store to a cache line is made visible to non-coherent agents by updating main memory with the value held in a dirty cache line.

A cache invalidate operation is used to ensure that a load from a location does not use a cached copy and therefore accesses main memory. This enables a store by a non-coherent agent, that cannot change a cached value, to be seen.

The following cache maintenance operations are supported:

- CleanShared. This transaction is used for a cache clean operation.
- CleanInvalid. This transaction is used for a cache clean and invalidate operation.
- MakeInvalid. This transaction is used for a cache invalidate operation.

Cache maintenance transactions differ from other snoop transactions because they can be required to propagate downstream to all caches. The cacheability attributes, signaled by **ARCACHE**, determine if downstream caches must observe a cache maintenance operation.

Cache maintenance transactions can be used in Non-shareable, Inner Shareable, and Outer Shareable domains. See *Shareability domain types* on page C3-158. The domain signaling **ARDOMAIN**, that accompanies a cache maintenance transaction, determines which hardware coherent peer caches must be snooped during the transaction.

Cache maintenance transactions are not permitted to use the System domain. However, all other **ARCACHE** and **ARDOMAIN** combinations that Table C3-3 on page C3-159 shows as permitted or legal are valid.

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